

CS615(C) Internet Solutions Engineering

WebApp Group Project

Catalogue

Each project will create a functioning website. As part of this, a project will develop a data-model (realisable in JSON), together with an accompanying RESTful API together with a prototype interface that demonstrates consumption of the API.

Project #1: Collaborative Story Creation Tool

(4-person project)

This project creates a website that facilitates collaborative creation of a story. You are responsible for implementing the following user stories:

- i. A user signs into their online account, edits an existing or creates a new story and saves this story in an online database. For now, a story has a title, a main text and contextual annotation in the form of tags.
- ii. A story can also have multiple co-called snapshots (organised as an ordered list) that represent selective key events in that story that are especially important (e.g., an assembly of people or a disastrous event). For now, the snapshot is represented as a text. Each of these snapshots can have multiple links to resources (e.g., to Wikipedia articles, pictures etc.) that are part of this snapshot.
- iii. Users may wish to collaborate in the creation and modification of stories. All users do have access to all stories for viewing and editing. However, a story can only be edited by one user at any time. If a user edits, others can only view.
- iv. A central log is kept that keeps record of all views and edits made with information about time, users and actions.

Projects #2: Restaurant Takeaway Site

This project will create a website that allows individuals to order takeaway food from a restaurant, and which allows the owner of the restaurant to manage the website. You are responsible for implementing the following user stories that define how the restaurant owner and customers interact with the system:

- i. The owner signs into their online account and can edit their list of delivery days (Monday – Sunday) and times; and the number of delivery slots that are available at each day-time.
- ii. The owner signs into their account and adds a food item to their list of foods available to order. They can also edit existing food items. For each food item a name, textual description, and price are included. By default, a food item's status is in 'draft' while later, when completed, then it is set to 'published'. Once published, it is publicly viewable on the website.
- iii. Customers sign into their online account and view the published menu. Customers, when viewing the menu, can select items to purchase, and choose available delivery date and time. The chosen delivery time is removed from the businesses list of delivery times for that date-time.

Projects #3: Collaborative Management of Tasks Tool

You are responsible for implementing the following user stories for a tool that allows for the collaborative management of tasks:

- i. A user signs into their online account, creates a task or edits a task previously created. For now, a task consists of a title, a description, and an optional link to a resource (e.g., to a website or an online image), and optional contextual annotations (e.g., tags that further describe the task, such as 'important', 'project X'). The user may decide to make the task private-only access, to share it with a group, or to make it publicly accessible. A user signs into their online account and may pick a task and, mark it as completed.
- ii. Any user may create and manage groups which can be used to share tasks within this group. Any user who creates a group automatically owns this group and can then also invite other users (or remove them).

Projects #4 - 7: Conference Management System

These projects will create a website to support conference management. Specifically, a scientific conference needs a system to manage paper submissions and the reviewing process. Researchers and practitioners can submit their scientific papers. There is a list of reviewers (program committee) who can review and evaluate the submitted papers. The conference chairs will distribute the papers to relevant reviewers. The reviewers then submit their reviews/comments and score to the system. The conference chair will make the decision about accepting the papers to be presented and published or rejecting the papers. The conference chair then notify the authors about the status of their submission.

#4: Conference Management System (1)

You are responsible for implementing the following user stories:

- i. An individual who will be an author or reviewer for the conference requests to set up an online account. To do this they complete a new author or reviewer account request. This consists of setting an account username and password; and creating a profile (with their name, affiliation, full affiliation address, an email address, a contact number, and a link that points to the website of the university/organization that they are affiliated with) and saves this information to an online database. The account when logged into then shows the message 'pending approval'.
- ii. An organizer of the conference (conference organizers have administrator rights), signs into their online account and sees the list of individuals who are pending approval. They can click to approve an account; at which time the account status changes from 'pending' to 'approved'. Once the account has been approved the individual can use their account.
- iii. When approval has been granted to an author, they can sign into their account and submit a publication. For this they enter: paper title, authors, keywords, abstract and pdf attachment. Once the paper details have been entered, they can select to 'submit' the paper for review. They also have the option to save a 'draft' for future editing.

Note that your group will have to liaise with the other groups implementing *Conference Management System projects* with respect to the data model to ensure compatibility and overall functionality.

#5: Conference Management System (2)

You are responsible for implementing the following user stories:

- i. An organizer of the conference (conference organizers have administrator rights), signs into their online account and sees the list of papers that have been submitted to the conference for review. For each paper they see: paper title, authors, keywords, abstract, pdf attachment, and name of

author that submitted the paper. For the name of author that submitted the paper they can also choose to view the profile details for the author (which are: their name, affiliation, full affiliation address, an email address, a contact number, and a link that points to the website of the university/organization that they are affiliated with).

- ii. An organizer can also view the list of Reviewers. For each reviewer they can also choose to view their profile details (which are: their name, affiliation, full affiliation address, an email address, a contact number, and a link that points to the website of the university/organization that they are affiliated with).
- iii. An organizer can assign papers to reviewers and view assignments made. For each paper 3 reviewers are assigned. The system automatically calculates the number of papers that should be given to each reviewer to review and uses this information to support the organizer in ensuring a fair distribution of the reviewing workload. A decreasing counter then ensures that each reviewer receives, as far as is possible, the same number of papers to review.

Note that your group will have to liaise with the other groups implementing *Conference Management System projects* with respect to the data model to ensure compatibility and overall functionality.

#6: Conference Management System (3)

You are responsible for implementing the following user stories:

- i. A reviewer of the conference logs into their online account and can view the list of papers assigned to them to review. For each paper they see: paper title, authors, keywords, abstract, pdf attachment, and name of author that submitted the paper.
- ii. The reviewer has the option to enter a review for a paper. Details to enter are: overall score (2 accept to -2 reject), review details, private comments to other reviewers and conference organizers. The reviewer can save their review in draft format for future editing. When the reviewer clicks submit on their review, they cannot make further edits to the review.
- iii. When a reviewer submits their review for a paper, they can see the other reviews already submitted for the paper. Their review is also visible to other reviewers who have already submitted a review; but hidden from those who have not yet submitted their review.

Note that your group will have to liaise with the other groups implementing *Conference Management System projects* with respect to the data model to ensure compatibility and overall functionality.

#7: Conference Management System (4)

You are responsible for implementing the following user stories:

- i. An organizer of the conference (conference organizers have administrator rights), signs into their online account and sees the review status of all papers. Details they can view for each paper are: reviewers name, overall score (2 accept to -2 reject), review details, private comments to other reviewers and conference organizers. They can choose to sort based on review score.
- ii. The organizer has the option to accept or reject a paper. It should also be possible to sort the papers based on the reject/accept status of papers. When accepting/rejecting a paper, the organizer also provides a brief metareview explaining their overall recommendation. This can be saved in draft format for future editing. During this process the paper review is hidden from authors. When the organizer clicks submit all reviews and the final accept/reject decision are visible to the paper author.

Note that your group will have to liaise with the other groups implementing *Conference Management System projects* with respect to the data model to ensure compatibility and overall functionality.

Projects #8 & 9: Theatre Booking System

These projects create a website for a theatre's online booking system.

#8: Theatre Booking System (1)

You are responsible for implementing the following two user stories that define how an admin uses the system:

- i. An admin signs into their online account and creates and/or edits list of entertainment shows currently running at the theatre (including show name, genre, brief description, dates, price). There is also a number of available tickets remaining associated with each date of the show. By default, during this process the edited entertainment shows are in 'draft' and not publicly viewable on the website.
- ii. An admin signs into their account and can sort and view the list of entertainment shows, and choose which shows to have in 'draft' status and which to have 'published' to the website. They can edit the list, as detailed in user story (i) above. When an admin is working with a show's details they are not editable by other admins.

Note that your group will have to liaise with the group that implements *Theatre Booking System (2)* with respect to the data model in order to ensure compatibility and overall functionality.

#9: Theatre Booking System (2)

You are responsible for implementing the following two user stories that define how an individual member of the public uses the system:

- i. Individual goes to the website and views list of entertainment shows available to book at the theatre, and can filter to view different genres of shows, different dates, availability of tickets and sort by price. For each show the following details are available: show name, genre, brief description, dates, price, number of available tickets. Note – for price there are multiple prices of tickets available.
- ii. Individuals can select a show to book their preferred price tickets for it. The chosen tickets are removed from the list of available tickets for that show on that date. The individual booking the ticket also enters their name, address and phone number when making their booking. If they do not complete the booking the tickets should be released back into the pool of available tickets.

Note that your group will have to liaise with the group that implements *Theatre Booking System (1)* with respect to the data model in order to ensure compatibility and overall functionality.